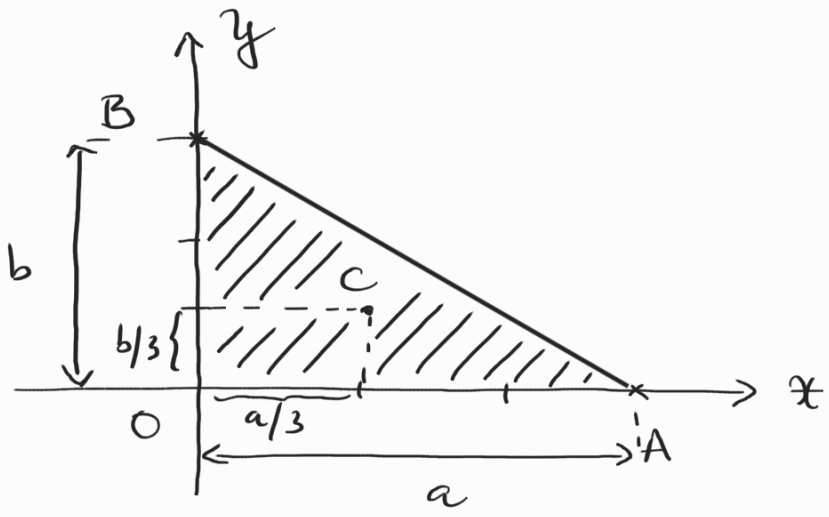
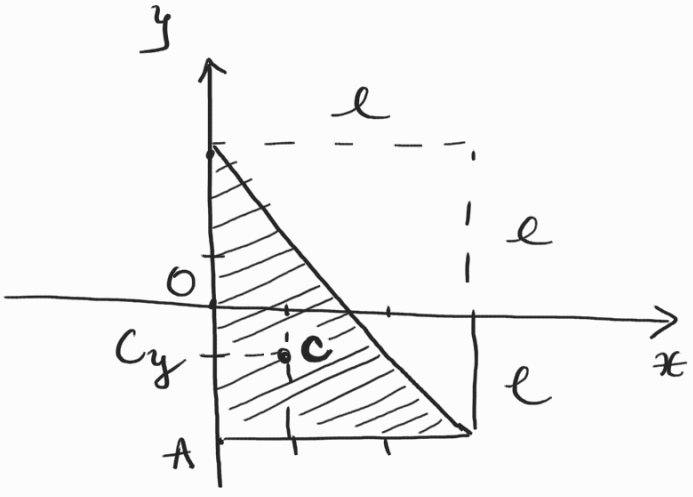
Centre de masă
PLĂCI OMOGENE

Triunghi dreptunghic



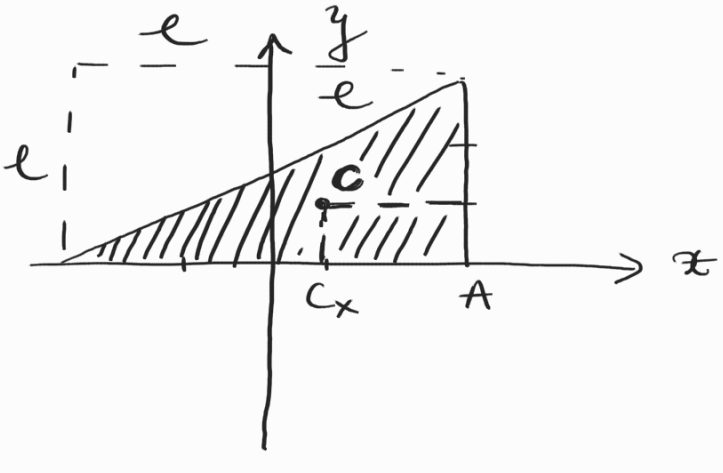
Centrul de masă se află la **o treime** față de **bază** și **2 treimi** față de **vârf**.



$$x_C = \frac{l}{3}$$

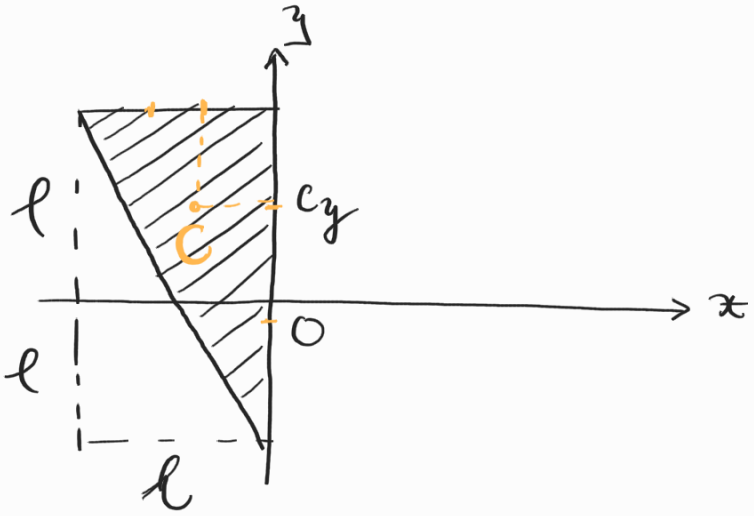
Coordonatele centrelor de masă se caută față de punctul O. Pentru y_C ne interesează distanța de la O la C_y. Pentru că centrul de masă C se află sub axa Ox, vom ține cont de semnul „−" pentru coordonata y_C

$$y_C = OC_y = -[OA - C_yA] = -\left[l - \frac{2l}{3}\right] = -\frac{l}{3}$$



$$y_C = \frac{l}{3}$$

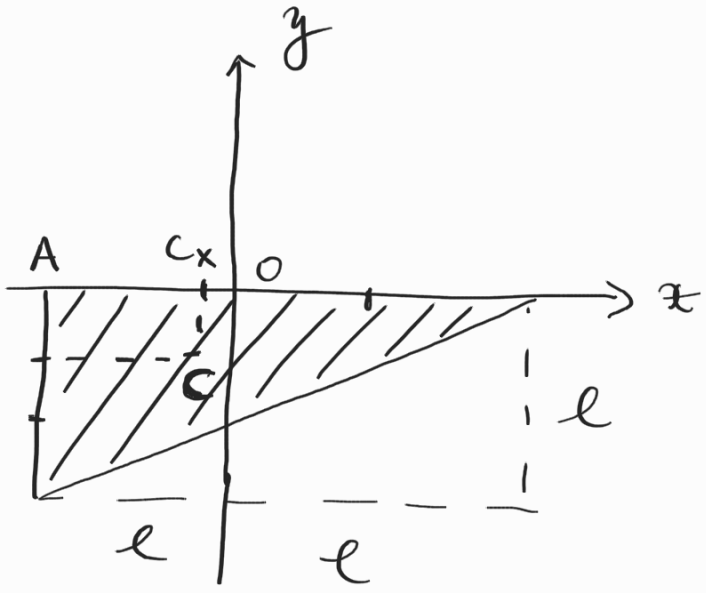
$$OC_x = x_C = OA - C_xA = l - \frac{2l}{3} = \frac{l}{3}$$



$$x_C = -\frac{\ell}{3}$$

$$y_C = OC_y = OA - C_yA =$$

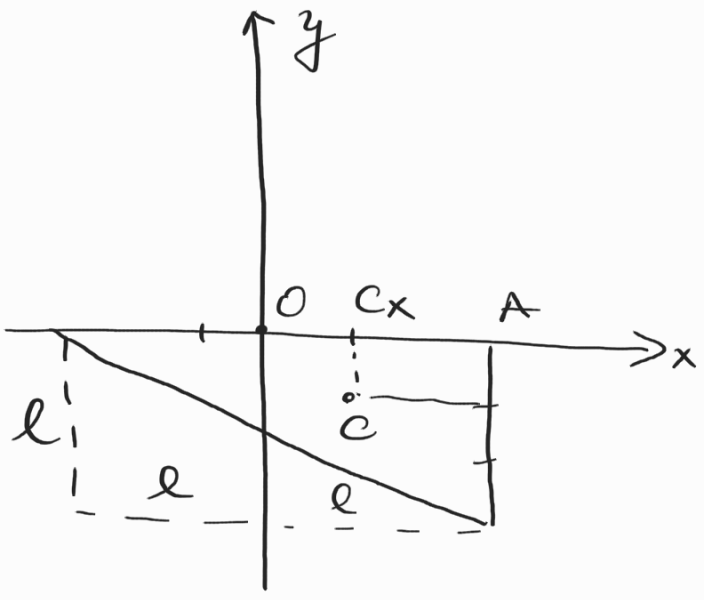
$$= \ell - \frac{2\ell}{3} = \frac{\ell}{3}$$



$$x_C = OC_x = -[OA - C_xA] =$$

$$= -\left[\ell - \frac{2\ell}{3}\right] = -\frac{\ell}{3}$$

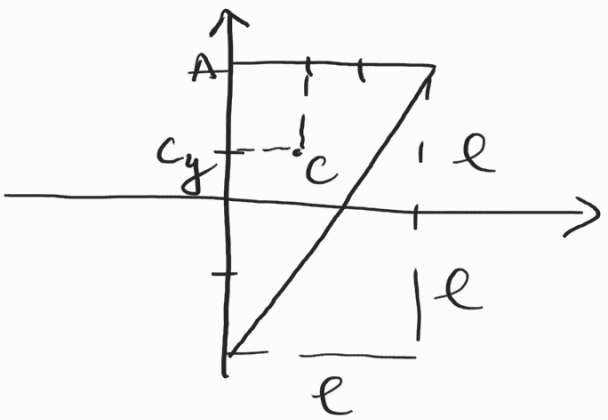
$$y_C = -\frac{\ell}{3}$$



$$OC_x = x_C = OA - C_xA =$$

$$= \ell - \frac{2\ell}{3} = \frac{\ell}{3}$$

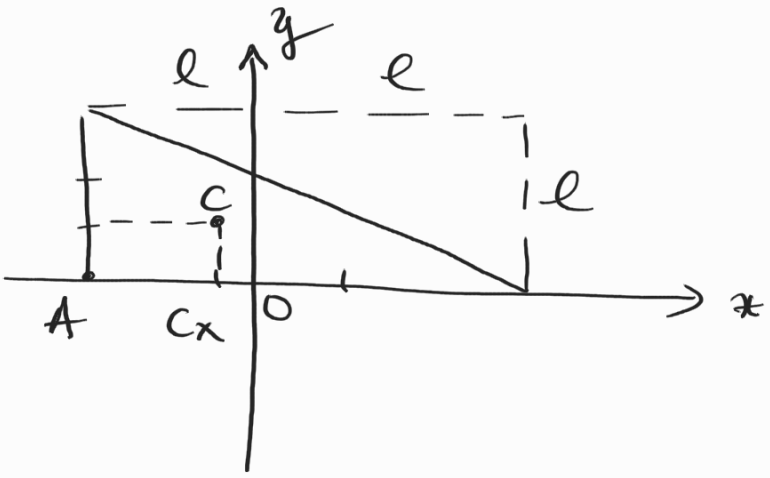
$$y_C = -\frac{\ell}{3}$$



$$x_C = \frac{\ell}{3}$$

$$OC_y = y_C = OA - C_yA =$$

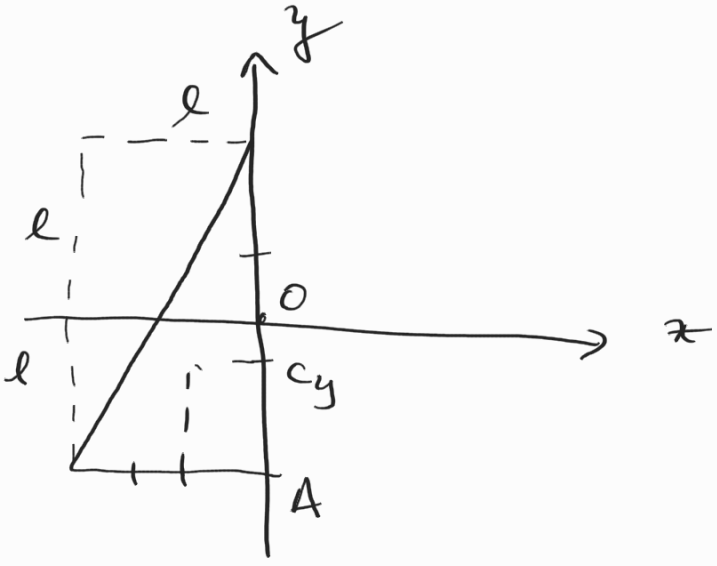
$$= \ell - \frac{2\ell}{3} = \frac{\ell}{3}$$



$$x_C = OC_x = -[OA - C_xA] =$$

$$= -\left[l - \frac{2l}{3}\right] = -\frac{l}{3}$$

$$y_C = \frac{l}{3}$$



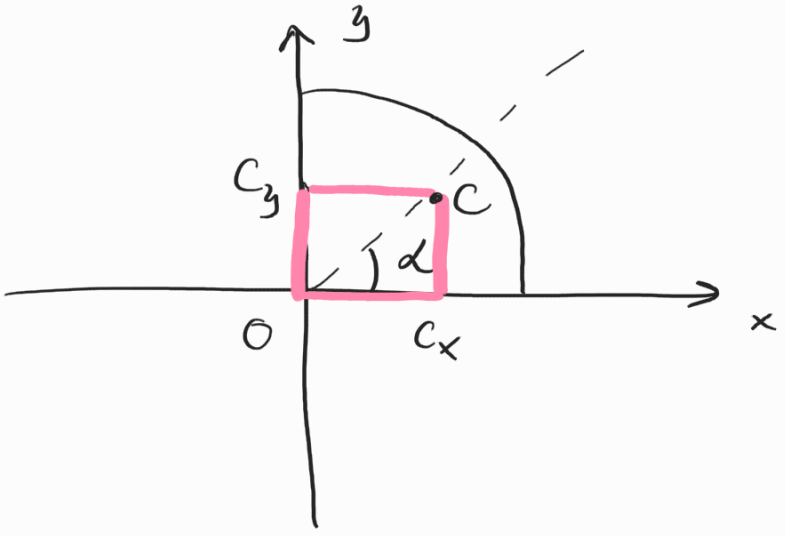
$$x_C = -\frac{l}{3}$$

$$y_C = OC_y = -[OA - C_yA] =$$

$$= -\left[l - \frac{2l}{3}\right] = -\frac{l}{3}$$

Centre de masă
PLĂCI OMOGENE

SFERT DE CERC (considerăm raza R)



Distanța OC se calculează cu formula

$$OC = \frac{2}{3}\,\frac{R\sin\frac{\pi}{4}}{\frac{\pi}{4}}$$

$$OC = \frac{2}{3}\,\frac{R\frac{\sqrt{2}}{2}}{\frac{\pi}{4}} =$$

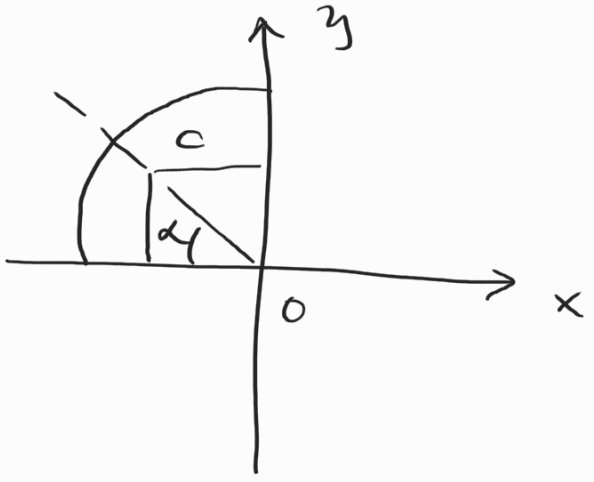
$$= \frac{8R}{3\pi}\cdot\frac{\sqrt{2}}{2} = \frac{4R\sqrt{2}}{3\pi}$$

Pentru a afla coordonatele centrului de masă proiectăm punctul C pe axele Ox și Oy.

În $\triangle OC_xC$ cunoaștem ipotenuza OC calculată mai sus și unghiul $\alpha = \frac{\pi}{4}$. Folosind funcțiile trigonometrice sin și cos calculăm coordonatele centrului de masă C.

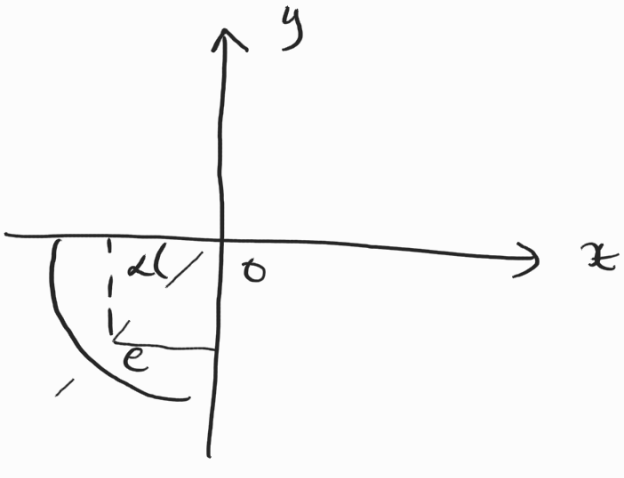
$$x_C = OC_x = OC \cdot \cos\alpha = OC \cdot \frac{\sqrt{2}}{2} = \frac{4R\sqrt{2}}{3\pi} \cdot \frac{\sqrt{2}}{2} = \frac{4R}{3\pi}$$

$$y_C = OC_y = OC \cdot \sin\alpha = \frac{4R\sqrt{2}}{3\pi} \cdot \frac{\sqrt{2}}{2} = \frac{4R}{3\pi}$$



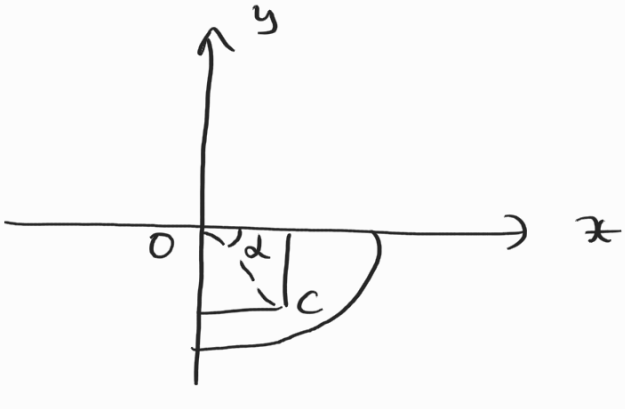
$$x_C = -OC \cdot \cos\frac{\pi}{4}$$

$$y_C = OC \cdot \sin\frac{\pi}{4}$$



$$x_C = -OC\cos\frac{\pi}{4}$$

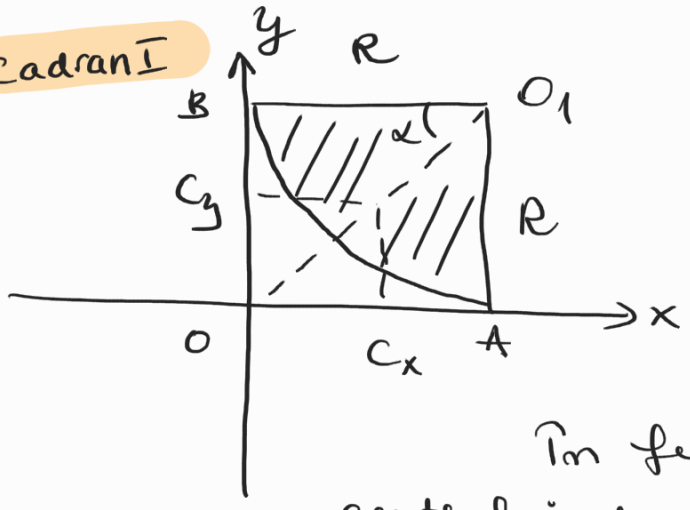
$$y_C = -OC\sin\frac{\pi}{4}$$



$$x_C = OC\cos\frac{\pi}{4}$$

$$y_C = -OC\sin\frac{\pi}{4}$$

Cadran I



Pentru a determina coordonatele centrului de masă C, ne interesează să calculăm distanțele OC_x și OC_y.

În funcție de poziționarea centrului de masă C în cadranele I-IV, vom ține cont de semnul „-".

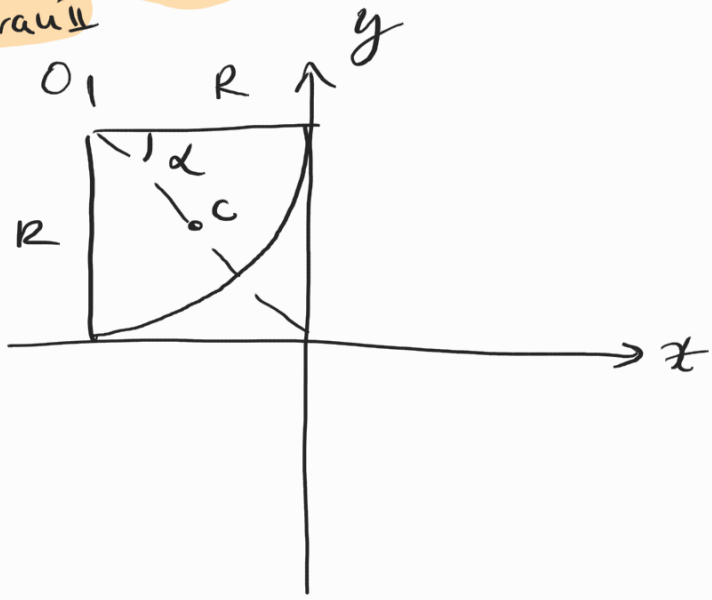
$$x_C = OC_x = OA - C_xA = R - O_1C \cdot \cos\alpha = R - \frac{4R\sqrt{2}}{3\pi} \cdot \frac{\sqrt{2}}{2}$$

$$= R - \frac{4R}{3\pi}$$

$$y_C = OC_y = OB - C_yB = R - O_1C \cdot \sin\alpha = R - \frac{4R\sqrt{2}}{3\pi} \cdot \frac{\sqrt{2}}{2}$$

$$= R - \frac{4R}{3\pi}$$

Cadran II



$$x_C = -[R - O_1C \cdot \cos\frac{\pi}{4}] =$$

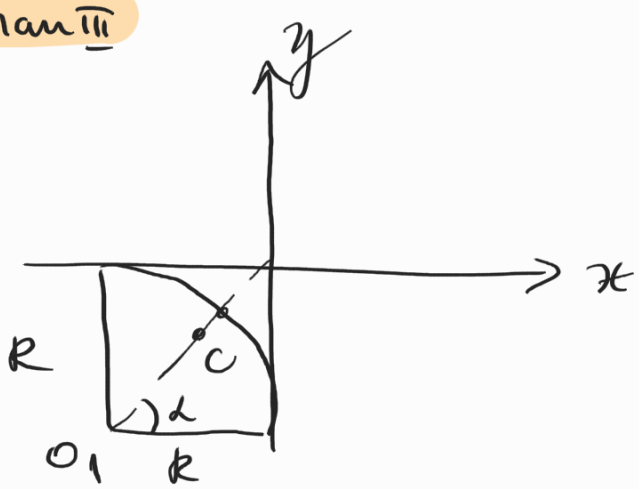
$$= -[R - \frac{4R\sqrt{2}}{3\pi} \cdot \frac{\sqrt{2}}{2}] =$$

$$= -R + \frac{4R}{3\pi}$$

$$y_C = R - O_1C \cdot \sin\frac{\pi}{4} =$$

$$= R - \frac{4R}{3\pi}$$

Cadran III



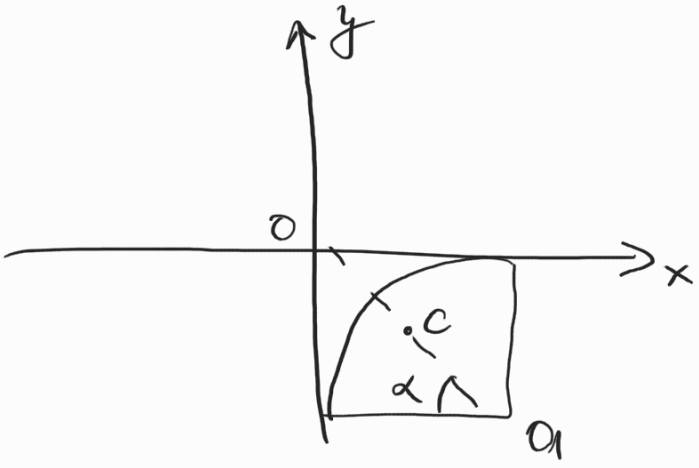
$$x_C = -[R - O_1C \cdot \cos\frac{\pi}{4}] =$$

$$= -R + \frac{4R}{3\pi}$$

$$y_C = -[R - O_1C \cdot \sin\frac{\pi}{4}] =$$

$$= -R + \frac{4R}{3\pi}$$

Cadran IV



$$x_C = R - O_1C \cdot \cos\frac{\pi}{4} = R - \frac{4R}{3\pi}$$

$$y_C = -\left[R - O_1C \cdot \sin\frac{\pi}{4}\right] = -R + \frac{4R}{3\pi}$$